Naomi Ikpera

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Education

Georgia Institute of Technology — Atlanta, GA
Bachelor of Science in Mechanical Engineering (GPA: 3.94/4.00) — *Aug. 2024 – Dec 2028*

Experience

Mechanical Design Researcher — Feb. 2026 – May 2026
RADAR/LIDAR Hardware-in-the-Loop Research Project — *Atlanta, GA*

- Co-led CAD design of a rig used to adjust the positioning of a RADAR sensor and study misalignment failures.
- Designed a compact sensor enclosure in SolidWorks to integrate TI AWR2243 radar, Intel RealSense, and Raspberry Pi components, incorporating 0.30-in mounting slots for controlled sensor misalignment testing.

Stairlift System Developer — Aug 2025 - Dec 2025
Stair Lift v2.0 – Engineering Design Project — *Atlanta, GA*

- Developed a mechanical mounting assembly for the stairlift carriage, considering a 300-lb design payload and load distribution across structural supports.
- Created CAD motion animations to demonstrate staircase folding and unfolding, evaluate component movement, and visualize mechanical interactions throughout operation.

Knack Academic Peer Tutor — Mar 2025 – Present
Georgia Tech — *Atlanta, GA*

- Tutored over 90 undergraduate students in General Chemistry, Calculus 1-3, Physics I, and introductory programming concepts, utilizing MATLAB simulation tools to demonstrate complex phenomena.
- Break down complex STEM topics using step-by-step explanations tailored to each student's learning style and reinforce concepts with hands-on examples.

College of Engineering Ambassador — Aug 2025 – Present
Georgia Tech — *Atlanta, GA*

- Represent the Dean of the College of Engineering at university events, information sessions, and outreach programs.
- Lead campus and engineering tours for prospective students, families, and adult visitors, communicating academic programs, facilities, and student life at Georgia Tech.

Projects

EchoDock | *SolidWorks, Surface Modeling, Parametric Design* — Dec. 2025

- Designed an ergonomic wireless earbud housing in SolidWorks using advanced surface modeling
- Developed an AirPod-inspired enclosure with a target wall thickness of 1.5 mm and 0.5 mm internal assembly clearances to accommodate electronic components

Disc Brake System | *SolidWorks, Mechanical Design, Parametric Modeling* — Dec. 2025

- Designed a 3D disc brake assembly in SolidWorks, modeling key components including the brake rotor, caliper, brake pads, pistons, and mounting brackets.
- Applied parametric modeling and assembly constraints to evaluate component alignment, rotor clearance, and overall mechanical functionality.

Competitive Task Robot | *Mechanical Design, Mechatronics, Failure Analysis* — Jan. 2026 – May 2026

- Designed a multi-arm rotary dispensing mechanism in SolidWorks to sequentially propel Lumas through controlled rotational motion, achieving a discharge velocity of 0.5 ft/s.
- Collaborated on the development of an autonomous robotic system integrating mechanical actuation, drivetrain components, and task-specific mechanisms within a $12 \times 24 \times 18$ in. design envelope.
- Engineered a rotary dispensing assembly featuring radially arranged arms and a circular housing to guide game pieces toward a designated discharge opening through controlled rotation.

Technical Skills

Programming: Python, Arduino, MATLAB
Software Tools: MS Office, Inkscape, SolidWorks, Ansys FEA
Machining & Fabrication: Lathe Operations, Milling, Waterjet Cutting
Engineering Concepts: CAD, Weldments, Prototyping, DFM, DFA, GD&T, Data Analysis, Continuous Improvement
Relevant Coursework: Stress Analysis, Torsion in Shafts, Dynamics of Bodies, Computing Techniques, Circuits and Electronics